Public App (by Inshorts): Delivering essential real-time video content to India's farthest reach with Media CDN

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Inshorts delighte
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implementation.

About Inshorts

Inshorts (https://www.inshorts.com/) is known for its namesake app which enables users to consume news in 60-word briefs and is very popular among India's urban millennials. In 2019, the company launched a new video platform Public, to bring videos that convey hyper-local content to tier 2, tier 3 and rural Indians coming online for the first time. In just 6 months, the Public grew to more than 10 million users. During the pandemic, its vital services enabled it to increase its user base 10-fold. Available in all languages in India, Public is today the nation's

Google Cloud results

- Boosts customer satisfaction by ending streaming delays that frustrated users on legacy CDN
- Enables optimized investment in app development through a 15% reduction in DevOps costs through the elimination of egress fees
- Frees engineers to focus on creative solution building by saving three hours per

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largest location-based tech platform, offering community and local updates, jobs and events information, and religious and cultural programs, at a hyper-local level. Inshorts has been named to the best app list of Google Play and is the winner of IAMAI's Best Innovative App award.

Industries: Media & Entertainment

Location: India

(https://cloud.google.com/), Media CDN

About Searce

Searce is one of the largest Google Cloud partners globally with over 12 years of experience, 150+ Google Cloud certifications and 300+ Google Cloud Experts in-house. It works with digital native companies and enterprises to re-imagine and futurify their business.

About Searce

Google Cloud (https://cloud.google.com/)

Media CDN (https://cloud.google.com/media-cdn)

BigQuery (https://cloud.google.com/bigquery/)

experience, 150+ Google Cloud

Google Cloud Armor (https://cloud.google.com/armor/)

day in mundane provisioning tasks
- Supports real-time tackling of data transmission issues with an intuitive dashboard that provides instant performance metrics

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India is a vast and vibrantly diverse subcontinent whose population of 1.38 billion is expected to become the world's largest (https://www.reuters.com/world/asia-pacific/india-set-overtake-china-worlds-most-populous-nation-2023-01-17/) in 2023. This massive market is united in its love of mobile content consumption, with a recent study (https://trak.in/tags/business/2021/02/12/indians-1-globally-in-spending-time-on-mobile-short-video-consumption-rise-by-4-times/) showing Indians spend more time on smartphones than any other country in the world. Leading the appetite is skyrocketing consumption of short video content, expected to soar four-fold by 2025.

Online content aggregator Inshorts (https://www.inshorts.com/) both taps and serves

India's passion for mobile video content with its hugely popular Public app, a video platform launched in 2019 that enables users to share community updates directly and in real-time at a hyper-local level. Demand for the app soared in the pandemic, rising 10-fold to 100 million registered users, as people sought accurate local information amid lockdowns and other restrictions that hampered traditional forms of media.

To carry out its mission, Inshorts needs to deliver vital local information, such as pandemic updates or news on fires or crime outbreaks, with the utmost speed in situations where lives can be at stake. That requires the most powerful content delivery network (CDN) available to minimize latencies, or transmission lags, in the streaming of user content. A CDN is an interconnected web of data centers that speeds up online data transmission by bringing content to the server closest to the end user.

For years, Inshorts relied on the CDN of one of the major global public clouds, only to find its network stability flagging as Public began enjoying exponential growth, reaching ever deeper into India's Tier-3 cities and rural hinterlands. Advised by partner Searce (https://www.searce.com/), Inshorts decided upon a bold switch to Media CDN (https://cloud.google.com/media-cdn) on Google Cloud (https://cloud.google.com/), attracted by the prospect of enjoying the same CDN infrastructure that YouTube deploys to serve global users.

"The results of our migration to Media CDN far exceeded our expectations, as we immediately gained a 10% improvement in latency, and unlocked unforeseen benefits in cost and operational efficiencies," Saurabh Jain, V.P Engineering at Inshorts. "That gave us the confidence to begin planning to roll out Public for global expansion."

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Going "all-in" with Google Cloud to serve Indian communities through network and operational transformation

Since its launch in 2013, as a mobile content aggregator that condenses news into articles of 60 words or less, Inshorts progressively moved from a multi-cloud environment to Google Cloud. The reason is that Inshorts began to find consistent and reliable cost, performance, and ease-of-use advantages with Google Cloud compared to its parallel service.

By 2020, Inshorts had made a near complete infrastructure migration to Google Cloud. The only cloud service that remained with the other cloud provider was its CDN, due to a perceived difficulty in switching delivery networks. After Inshorts launched Public, however, it found itself increasingly affected by video transmission rebuffers that frustrated users on its legacy CDN.

This continued to be an issue as Public's popularity soared in the Covid crisis, thanks to the timely community information it provided. Meanwhile, Inshorts found itself struggling with the time-consuming and costly DevOps provisioning required to move video data out of Google Cloud into the rival provider's CDN.

These issues converged to cause Inshorts to go "all-in" with Google Cloud in 2020 by switching to Media CDN. A key motivator was learning from Searce that Google Cloud offered an unrivaled network of points of presence (PoPs) or strategic data centers that communicate with end users in a geographic area, in 200 countries and more than 1,300 cities. Inshorts also learned from Searce about the significant operational and cost advantages it would gain by integrating CDN into its cloud architecture. This is a benefit that was amplified by the API-first approach of Media CDN, which automates the provisioning of data pipelines for content release, in a serverless environment.

Ultimately, Jain says, any concerns about a difficult CDN migration evaporated overnight.

"Working with Searce, we were surprised by the ease with which we carried out a seamless migration to Media CDN in under two days, making Google Cloud our sole cloud provider," he recalls. "It required only the switching of the plug between the two services."

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Eliminating egress costs to power the next stages of platform evolution

One of the most powerful advantages Inshorts has gained by moving to Media CDN is the elimination of egress costs from its Google Cloud architecture to the legacy provider's CDN, which enables investment in strategic planning and solution building. Egress costs are network fees that cloud providers charge to move data out of their cloud, and the costs can be substantial. By opting for a 100% Google Cloud infrastructure, Inshorts has been able to create a direct network pipeline between Google Cloud

Storage, where it stores its Public video data, and Media CDN. Jain says that results in 15% cost savings on overall DevOps costs, thanks mostly to the elimination of egress fees, alongside optimized, near-zero latency content delivery to end users.

Meanwhile, since all cloud services are now with Google Cloud, the DevOps team is freed from the mundane yet complicated tasks of provisioning network links between different cloud environments. Jain says that spares each engineer up to three hours per day in workload. A further advantage comes from the intuitive dashboard of Media CDN, which gives a clear, real-time view of performance metrics and monitoring. That means the team can address any data transmission issues as quickly as they arise, anywhere in a now fully integrated Google Cloud stack.

Taking the hyperlocal vision global with powerful data analytics and top-flight security

Now that Inshorts has touched lives in the farthest reaches of India, it plans to take Public to a global audience. To build a secure international network, Inshorts is working with Searce on a Google Cloud Armor (https://cloud.google.com/armor) implementation to ensure that its platform is safe from denial of service (DDoS) attacks that could interrupt services when it's needed most, such as during natural disasters and other emergencies.

The platform is also working with Searce on the adoption of BigQuery (https://cloud.google.com/bigquery) to upgrade its data analytics capabilities for generating performance reports on everything from content popularity to system speed and stability. Up to now, data analytics reports have been a manual process in which developers have needed to provision internal servers to run data streams through Inshorts algorithms. With BigQuery, Jain says, Inshorts will gain an automated, planet-scale data analytics solution, expected to reduce performance report generation from three days to one.

Searce is assisting in this operation by executing a data compression program for all Inshorts data that runs through BigQuery. This promises to generate significant time and cost savings in deploying the auto-scaling data warehouse.

"Since going 'all-in' with Google Cloud as our sole cloud provider, we've opened up an unlimited world of potential to transform our solution for a global audience," says Jain. "Adopting Media CDN has done more than bring us unrivaled video content transmission speeds. It has unlocked new paths to creative capabilities."

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